

$$\begin{aligned}
& \mathbf{J} \times \mathbf{H} \\
& \mathbf{E} \mathbf{H} \\
& \mathbf{i} \\
& \mathbf{S} \\
& \mathbf{L} \\
& (r,\phi,z) \\
& e^{i\ell\phi} \\
& = \\
& \hbar \\
& L_z \\
& \ell \\
& \hbar \\
& \partial E_x/\partial x \\
& \partial E_y/\partial y \\
& \nabla \cdot \\
& \mathbf{E} = \\
& \frac{\partial E_x}{\partial x} + \\
& \frac{\partial E_y}{\partial y} + \\
& \frac{\partial E_z}{\partial z} = \\
& 0 \\
& E_z, E_y \\
& s_{\pm 1} \\
& s_{\pm 1} \\
& +1 = \\
& \mathbf{E}_0(r,z)(\hat{x} + \\
& is\hat{y})e^{ikz} \\
& \propto \\
& (\hat{x} + \\
& is\hat{y}) = \\
& (\cos\phi\hat{\rho} - \\
& \sin\phi\hat{\phi}) + \\
& is(\sin\phi\hat{\rho} + \\
& \cos\phi\hat{\phi}) \\
& \theta \\
& \phi \\
& \hat{z} \\
& \hat{\rho} \\
& \hat{z} \\
& \hat{\rho} \\
& \propto \\
& \cos\phi + \\
& is\sin\phi \\
& \cos\phi + \\
& is\sin\phi = \\
& e^{is\phi} \\
& \theta \\
& \phi_p \\
& E_z \propto \\
& \int_0^{2\pi} (\cos\phi_p + \\
& is\sin\phi_p)e^{ik\rho\cos(\phi_p-\phi)}d\phi_p \\
& \int_0^{2\pi} e^{is\phi_p}e^{ik\rho\cos(\phi_p-\phi)}d\phi_p = \\
& e^{is\phi}\int_0^{2\pi} e^{is\alpha}e^{ik\rho\cos\alpha}d\alpha \\
& e^{is\phi} \\
& J_1(k\rho\sin\theta) \\
& \hat{z} = \\
& \hat{0} = \\
& \hat{z} = \\
& (\rho,\phi,z) = \\
& -iAe^{is\phi}I_1(\rho,z) \\
& A \\
& I_1(\rho,z) \\
& NA = \\
& n\sin\alpha \\
& {}_1(\rho,z) = \\
& \int_0^\alpha \cos^{1/2}\theta\sin^2\theta P(\theta)J_1(k\rho\sin\theta)e^{ikz\cos\theta}d\theta \\
& \theta \\
& \alpha \\
& P(\theta) \\
& J_1 \\
& E_z \\
& e^{is\phi} \\
& \ell \equiv \\
& s_{\pm 1} \\
& s_{\pm 1} \\
& +1 = \\
& \ell = \\
& +1 \\
& \hat{z} \\
& \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \phi^2} + 2ik\frac{\partial}{\partial z}\right)u(r,\phi,z) = \\
& 0 \\
& \exp(i\ell\phi) \\
& 2\pi \\
& \ell =
\end{aligned}$$